

Spectrum Access for the Passive Services: The Past and the Future

Passive (nontransmitting) users of the radio spectrum, mainly scientific users like radio astronomy and remote sensing, face particular challenges in accessing the radio spectrum. The author looks at the past and discusses challenges facing passive users in the future.

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ABSTRACT | The radio astronomy service (RAS), Earth exploration satellite service (passive) [EESS (passive)], and space research service (passive) [SRS (passive)], collectively known as the passive services, have unique spectrum requirements. Radio astronomy requires access to the spectrum at approximately octave intervals for continuum observations, and in addition to a large number of spectral lines that provide unique information about specific physical processes. The EESS, known more commonly as satellite remote sensing, has somewhat similar requirements. Due to the similarities, and because the passive services are much more sensitive to interference than active communication services, they have been mostly co-allocated and some bands have been allocated to them on an exclusive, primary basis. When looked at closely, however, the RAS and EESS (passive) requirements differ in terms of geographic location, timing, and other details. As the radio spectrum becomes increasingly congested, sharing on a band-by-band basis between active and passive services becomes increasingly difficult. This paper discusses how sharing between the passive services, particularly radio astronomy, and the active services evolved, and some of the issues the RAS and EESS community must resolve.

KEYWORDS | Radio astronomy; radio spectrum management; remote sensing

I. INTRODUCTION

The radio astronomy service (RAS), Earth exploration satellite service (passive) [EESS (passive)], and space research service (passive) [SRS (passive)] are collectively known as the passive services. These services have unique and, in some ways, similar spectrum requirements. Radio astronomy from space is conducted mostly under the SRS (passive) and will not be discussed separately in this paper.

All passive services need access to the spectrum for continuum as well as spectral line observations. The continuum spectrum is used by radio astronomers to define the variation of the radiation with frequency in sufficient detail to determine the physical mechanism responsible for the emission. It is also used to determine the electron density, the magnetic field, and other physical parameters in the source. To determine it well enough, access to the spectrum is needed at approximately octave intervals. The EESS (remote sensing) also uses continuum observations to determine the processes that give rise to the emission, as well as to separate the various processes that may give rise to the emission over a wide range in frequency from the soil, ocean surface, or other object of interest. The exact frequency at which an observation is made is not important for continuum observations. In addition to the continuum radiation, radio astronomers and remote sensing scientists need to observe spectral lines at specific frequencies that provide a wealth of information about the chemical composition, bulk and differential movements, magnetic field, and some other parameters in stars, galaxies, and the interstellar medium, as well as in solar system bodies.

Some of the techniques used by the EESS are similar to those used by radio astronomers, although there are important differences. For example, ground-based radio

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astronomers are not limited by the size or weight of their instruments and have developed some of the largest and heaviest instruments in existence [e.g., the Green Bank Telescope (GBT), at Green Bank, WV, USA, is the largest movable structure on Earth], a luxury that is not available to EESS practitioners, who must work with airborne and space-based instruments. All passive services strive to observe extremely faint signals, and are highly susceptible to interference, much more so than active communication services. Because of these similarities, the passive services have been mostly coallocated, with a number of bands allocated to them on an exclusive, passive basis. There are, however, differences in how the services operate that are likely to have an impact on their future.

II. PASSIVE SERVICES AND REGULATIONS: THE LAST CENTURY

Three stages can be discerned in radio astronomers' frequency protection efforts since Karl Jansky discovered radio waves of cosmic origin [1]. During the first of these stages, until the early 1980s, when nongeostationary satellite systems began to operate, interference protection of radio astronomy installations was relatively simple. The radio telescope, most often a large, single dish antenna, needed to be protected from interference that could be caused by a few stationary or, possibly, moving sources transmitting either cofrequency with the observed band or in a neighboring one. The first line of defense was, rather obviously, to locate the observatory as far as possible from sources of man-made radio transmissions. The best known example of this kind of protection is that of the telescopes within the U.S. National Radio Quiet Zone (NRQZ), established by the U.S. Federal Communication Commission (FCC) in 1958 [2]. The NRQZ houses today one of the best protected radio telescopes in the world at Green Bank, WV, USA: the GBT.

During these five decades radio astronomers worked hard to become integrated into the elaborate national and international regulatory structure that exists for the protection of all radio services, not just radio astronomy, and established a presence at the international as well as the national regulatory agencies. Radio astronomy was recognized as a "radio service" (although, quite appropriately, not a "radiocommunication service") at the 1959 World Administrative Radio Conference (WARC-59, Geneva, Switzerland). At WARC-59, the 1400–1427-MHz band, used to observe the neutral hydrogen atom that constitutes over 90% of the matter in the Universe, was allocated to the RAS and later to all passive services [3]. As radio technology progressed to higher frequencies, a number of other bands were allocated to the RAS at successive World Administrative Radio Conferences (WARCs) held by the International Telecommunication Union (ITU). As a final step in this process, a substantial fraction of the 70–275-GHz spectrum was reallocated to radio astronomy and to the other passive

services at the 2000 World Radiocommunication Conference (WRC-00, Istanbul, Turkey). Taking advantage of the high atmospheric absorption, most spectrum above 70 GHz was reallocated on a shared basis to both passive and active services.

In bands allocated to the RAS on an exclusive or shared primary basis, radio telescopes are, in principle, protected from cochannel transmissions of other licensed services by national and international regulations. The ITU radio regulations (RRs) [4] provide for registration of radio telescopes, so that interference in radio astronomy bands can be minimized at radio astronomy sites. The ITU–Radio Communication Sector (ITU–R) recommendations provide information about the level of protection needed [5], about the astrophysically most important frequency bands [6], and some other matters of importance to the RAS. During this period, due to the remoteness of most observatories, observing outside radio astronomy bands was possible most of the time, even in bands without any regulatory protection. Article 4.4 of the RRs provides explicitly for the use of frequencies that are not allocated to a service, as long as no harmful interference is caused to allocated services or claim protection from them. In fact, the RRs even allow the notification of radio astronomy stations that observe in unallocated bands:

“Any frequency to be used for reception by a particular radio astronomy station may be notified if it is desired that such data be included in the Master Register” (RR 11.12).

The RRs make it clear, however, that use of an unallocated band is on a noninterference, unprotected basis:

“The status of the radio astronomy service in the various frequency bands is specified in the Table of Frequency Allocations (Art. 5). Administrations shall provide protection from interference to stations in the radio astronomy service in accordance with the status of this service in those bands” (RR 29.8).

Radio astronomers took advantage of this *laissez faire* clause of the RRs. Passive observations do not cause interference, and when interference was received, bad data could be thrown out. At isolated locations, data contaminated by interference usually comprised a small fraction of the observations.

A second period in frequency protection efforts of the RAS goes from the early 1980s to about 2000. During this period, the efforts of the radio astronomy community were directed mainly toward obtaining protection from satellite emissions, particularly from their unwanted emissions. Geostationary satellites were first launched in the mid to late 1960s. Astronomical observations at and near the

vicinity of the geostationary orbit were made difficult by these satellites in some bands. For the most part this problem could be overcome by observing from telescopes located at various latitudes, making use of the parallax so obtained [7]. Deployment of nongeostationary (NGSO) satellite systems, e.g., GLONASS or GPS, began in the early 1980s. NGSO satellites orbit at heights ranging from a few hundreds to over ten thousand kilometers above Earth. They move quickly in and out of the sidelobes of radio telescopes, and each satellite may have multiple beams. Most NGSO systems provide coverage everywhere on Earth at all times by design, and for the first time denied the access of radio astronomers to some bands permanently. Radio astronomers resigned themselves to the loss of some bands, but wanted to retain access to the rest of the spectrum. Unwanted emissions by some NGSO systems made highly sensitive observations difficult or impossible even when those observations were being conducted at frequencies far removed from the band intentionally occupied by satellite transmissions, and sometimes even in passive or primary radio astronomy bands. While radio telescopes continued to be located at remote locations to avoid man-made interference as much as possible, remoteness did not afford protection any longer from interference by NGSO systems. Throwing out bad data was no longer an option either, as NGSO satellites interfered with observations at all times. In response to these developments, radio astronomers did attempt to secure more stringent regulations during the 1980s and 1990s. They tried to push new or existing satellite downlink allocations as far as possible from radio astronomy bands. Administrations, however, were reluctant to adopt this as a guiding principle for allocations. The effort was also complicated by the fact that astronomers are interested in many parts of the spectrum. Passive service advocates also tried to place mandatory regulatory limits on unwanted emissions from satellite transmitters. Some general rules applicable to space stations were adopted, and spurious emissions were restricted, but the outcome is far short of the levels required to protect the RAS or EESS bands. Nevertheless, progress was made, some of which is likely to benefit not only the RAS but also all radio services.

- Spurious emission limits were adopted, where none existed above 17 GHz before.
- Mandatory out-of-band emission limits were adopted for the protection of some radio astronomy bands from unwanted emissions of satellite systems that operate nearby in frequency. The limits are contained in footnotes to the ITU Table of Allocations.
- The 2003 World Radiocommunication Conference (WRC-03, Geneva Switzerland) adopted Res. 739 that contains guidance to Administrations on how to resolve problems arising between the RAS and satellite services due to unwanted emissions. The Resolution lists threshold levels that can be used to trigger consultations between Administrations.
- ITU-R Rec. RA.1513 [8] quantified the percentage of acceptable data loss to radio astronomy due to interference from other services.
- Recommendations were developed to calculate threshold levels of interference by NGSO systems through the adoption of the equivalent power flux density method.

EESS missions were first launched in the late 1960s [9], much later than the beginnings of radio astronomy. Since EESS missions look down, usually from low earth orbit rather than up from the ground-like radio telescopes, the source of the interference they received was mostly from ground-based, rather than satellite-based, transmitters. Most interference was generated by transmitters operating outside the EESS band of interest, particularly by radars. EESS practitioners tried to gain protection from unwanted emissions through regulations, and on occasions succeeded in doing so. For example, they gained limited protection from unwanted emissions of transmitters operating in bands adjacent to some of the passive bands. In particular, unwanted emissions of some services operating in the 1350–1400-MHz, 1427–1452-MHz, 22.55–23.55-GHz, 30–31.3-GHz, 49.7–50.2-GHz, 50.4–50.9-GHz, 51.4–52.6-GHz, 81–86-GHz, and 92–94-GHz bands were limited at WRC-03. Direct radio interference from satellites is a lesser concern for the EESS than for the RAS; however, reflection of satellite transmissions from the surface of the ocean, lakes, and wet soil presents serious problems.

III. THE FUTURE IS NOW

The third stage of radio astronomy frequency protection efforts began around 2000, and extends into the foreseeable future. It is marked by the enormous proliferation of low-power, unlicensed devices of all kinds. Mobile devices, of increased sophistication and data throughput requirements, are a huge part of this increase. These devices can operate under few restrictions in most bands. Some exceptions are the purely passive bands listed in footnote 5.340 of the RR that prohibits all emissions and the bands in which the GPS system and other global navigation satellite systems operate. More and more such devices are expected to come on the market and will require more spectrum. For example, the GSM coalition estimates that an additional 1.8 GHz of spectrum will be needed by 2020 to satisfy the demand for mobile broadband devices worldwide. Propagation characteristics require this demand be met in the 100-MHz–6-GHz range, approximately. The same frequency range will have to accommodate most present users as well. The problem for radio astronomers (and, fair to say, many other licensed radio services as well), is compounded by the fact that many of the mobile devices can switch between several bands and some employ ultrawideband (UWB) technology. UWB signals are low power in comparison with traditional communication systems, but are extremely broadband, and may generate 1-GHz-wide or even

wider signals. There is also a huge increase in the number of incidental or unintentional radiators, cameras, computers, etc., using digital technology that generates low-level radiation over a very broad range of frequencies. As a rule, radio astronomers do not have control over the emissions of devices beyond the observatory, so that interference from these types of devices is expected to increase.

Mobile broadband devices occupy preferentially the range of frequencies below 6 GHz, but are also beginning to appear much higher in the spectrum. For example, the 76–81-GHz band is beginning to be used by vehicular radars. Such radars can already be found in some newer model vehicles, and the automobile industry expects them to become standard equipment on all cars by 2020. Anticipation of many millions of cars outfitted with several radars in this band that are likely to cause interference to millimeter-wave telescopes not only in the 76–81-GHz band, but possibly also at its harmonics is already becoming a big concern for millimeter-wave astronomers.

The wireless revolution is as much a problem for the EESS, as it is for radio astronomy, perhaps even more so. Most EESS missions cover the entire surface of Earth, and cannot suffer data losses in excess of a fraction of a percent. Unlicensed transmitters not only cause interference that is extremely difficult to identify due to the mobile nature of the interferor, but they also may raise the noise floor over much of the surface of Earth, making EESS measurements much more difficult.

As noted already, radio astronomers always observed outside bands allocated to the service, among other reasons because those bands are too narrow, and sensitive observations would require very lengthy observations if confined to them. Increasingly, the science requires access to most of the spectrum for several reasons, and confining observations to RAS bands is not an option.

Interest in the study of the structure and evolution of the early Universe, and consequently in highly red-shifted spectral lines, increased enormously in the last couple of decades. Radio astronomy allocations cover a very limited range of Doppler shifts, particularly at centimeter- and decimeter-wavelengths. Even in some of the widest primary allocations, the band allocated to the RAS covers only a fraction of what is needed. For example, the 1420-MHz neutral hydrogen (HI) line has been observed red-shifted down to 500 MHz, a shift that represents a look into the Universe as it was approximately ten billion years ago. Searches for even higher red-shifts are under way. Going further back to 13 billion years or more would require observing the line in the 100–200-MHz range. In addition to neutral hydrogen, some of the most abundant molecular lines have been detected in external galaxies with velocities up to 50 000 km/s, corresponding to a frequency reduction of up to 17%. Observations in allocated RAS bands are clearly not a viable option for these studies.

Another reason for the desire to observe outside allocated bands is that spectral lines discovered during the

Table 1 A Few Spectral Lines of Astrophysical Importance and Their Allocation Status

Substance	Rest Frequency	Suggested minimum band	Notes
Methanol (CH₃OH)	6 668.518 MHz	6 661.8- 6 675.2 MHz	1
Helium (³He⁺)	8 665.65 MHz	8 657.0- 8 674.3 MHz	2
Methanol (CH₃OH)	12.178 GHz	12.17-12.19 GHz	2

last 30 years at frequencies below 60 GHz are not covered by any radio astronomy allocation, and most are not even mentioned in the RR. The existence of some important lines, for example, the 327-MHz deuterium line, has been long predicted, but a confirmed detection was made only in the last decade. Table 1 is a small extract from Rec. ITU-R RA. 314-10 (Preferred Frequency Bands for Radio Astronomical Measurements) that illustrates the point. The notes column of the table indicates cases when the width of the band allocated to radio astronomy is less than the minimum recommended, (and extends to a distance far closer in the Universe than astronomers would like to observe) (Note 1). In a couple of instances, the line is not even mentioned in the RR (Note 2). These are just a few examples; most astrophysically important lines must be observed outside an RAS allocation.

Finally, observing outside allocated bands is necessary because frontline science is linked to observations of the faintest sources. Since the sensitivity of an observation is proportional to the square root of the observed bandwidth, up to 10% fractional bandwidths are needed for observations of the faintest sources. In response to these requirements, modern radio telescopes can cover very broad frequency ranges and are outfitted with broadband receivers, regardless of frequency allocations. The tendency is even more conspicuous for the generation of radio telescopes now coming on line or in planning, e.g., the Atacama Large Millimeter–submillimeter Array (ALMA), at the Llano de Chajnantor, Chile or the Square Kilometer Array (SKA), in Western Australia and New Zealand (low-frequency array) and in the Karoo region of South Africa (medium- and high-frequency array). Interferometers that are increasingly used because of their high resolution are less susceptible to interference than single dish telescopes. Table 2 shows the frequency coverage of some of the new telescopes.

The comments column shows the location, the sponsoring country (or countries), if the instrument is a single dish telescope (S) or an interferometer (I), and where it is located.

Obviously, a telescope that covers a broad frequency range and is outfitted with the necessary broadband receiver

Table 2 Frequency Coverage of Some Newer Telescopes

Telescope	Band covered	Comments
LMT (Large Millimeter Telescope)	70-350 GHz	Mexico-USA collaboration – S (Baja California, Mexico)
Sardinia Radio Telescope	0.3-100 GHz	Italy- S (Sardinia, Italy)
eVLA (extended Very Large Array)	1-50 GHz	USA – I (New Mexico, USA)
ALMA (Atacama Large Mm-submm Telescope)	67-950 GHz	USA-Europe-Japan – I (Llano de Chajnantor, Chile)
LOFAR (Low Frequency Array)	30-240 MHz	Europe – I (The Netherlands and various European countries)

is not sufficient to carry out radio astronomy observations anywhere in the spectrum. In a crowded radio-frequency environment, this would not be possible without additional measures. For example, the low-frequency array (LOFAR), with stations located in northern Europe, covers the high-frequency (HF) and very-high-frequency (VHF) bands. To be able to observe, the instrument uses mitigation methods developed by the radio astronomy community for various situations. A comprehensive and up-to-date description of mitigation techniques used in radio astronomy can be found in ITU-R Report RA.2126 [10]. Though mitigation techniques make observations possible in some bands, this usually comes at the cost of loss of sensitivity. It should also be kept in mind that mitigation is applicable only in some special situations.

In contrast with the RAS service, there does not appear to be any need for the EESS to cover the whole spectrum. While the EESS community would need some additional allocations, for example, in the 6–7-GHz range [9], regulatory efforts of the remote sensing community appear to be oriented more toward controlling interference than toward new allocations [this statement does not apply to the EESS (active) that is not discussed in this paper]. As noted already, EESS interests successfully advocated limits to unwanted emissions into some of the passive bands used by the service at WRC-03. In some of the bands covered, there are mandatory unwanted emission levels applicable to some of the active services that operate in neighboring bands, while in others the levels are recommended. Space agencies that fly most EESS missions have often been successful in tracking down the sources of interference worldwide that operate in passive bands in violation of the regulations. They also succeeded on several occasions in requesting Administrations to eliminate such sources of interference.

There is an important difference between the RAS and the EESS that needs to be taken into account. Radio astronomers need to control interference at the telescope or,

in ITU language, at the site of the radio astronomy station. To reduce the level of emissions below a certain threshold at the telescope site, emissions need to be controlled within an area that may, in some cases, be very large. Still, geographical separation is possible between RAS stations and the active services at some point. It is unlikely, for example, that radio astronomers would ever make use of the 1400–1427-MHz band within the largest metropolitan areas, e.g., New York, Paris, or Los Angeles. On the other hand, most EESS missions must cover the entire surface of Earth, and geographical sharing in their case is not possible.

IV. CONCLUSION

The radio astronomy community needs to initiate a debate on how it views the future of the science in a changing and increasingly crowded radio spectrum environment. This debate must touch upon the usefulness of the regulations and on how the community participates in national and international regulatory fora. Some of the issues that should be discussed are listed as follows.

- Should radio astronomers continue to maintain that all relatively narrow bands allocated to radio astronomy below about 70 GHz are needed and are useful?
- Should the radio astronomy community attempt to trade protection in many bands at all radio astronomy sites for a high level of protection across most of the spectrum at a few places worldwide? This question is particularly important and timely since forefront research is done mostly using a few mega-instruments that are accessible to the international community on the basis of scientific merit.
- Should the Rec. ITU-R RA.769 levels continue to be used as the cornerstone for the protection of the next generation of instruments? Many of the new instruments are interferometers and since interferometers are less susceptible to interference than single dish telescopes in some instances the threshold levels of interference in Rec. ITU-R RA. 769 may be overprotective. On the other hand, while the recommendation assumes an observing time of 2000 s for a typical astronomical observation, observations of faint sources do require much longer integration times. If adopted as the standard observing time, longer integration times would require more stringent levels of protection that would be difficult to get recognized by most countries.
- How should the radio astronomy community attempt to influence the evolution of the regulations? For example, should instrument-specific levels that take into account the geographical outlay and frequency coverage of an instrument be calculated on a case-by-case basis?

- How can radio telescopes develop/take advantage of appropriate mitigation techniques and what are the limitations of those techniques?
- How can dynamic, cognitive methods be taken advantage of to benefit passive users of the spectrum, and how can such techniques be reflected in the regulations?

The usefulness of the EESS allocations, and the continued need for them, is not in question. Most of the problems of the EESS community seem to be related to better

interference protection on bands already allocated to the service. Once the radio astronomy community has come to an agreement on its own needs, the RAS and EESS communities should consider the continued usefulness of joint allocations.

- Do such joint allocations still respond to the requirements of both communities?
- If that is not the case, how to separate them?

A thorough debate of these questions is needed before changes in the existing regulations, if any, are adopted. ■

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